10/16

IDS detects attacks by observing network communication records.

Subsequently, labeled examples of **new attack** types **must be incorporated into the detector**.

→ Experiments to determine whether the system can be updated using **minimal costs and samples**, while **maintaining the detection quality for existing attacks**.

Attacker	Victim	Attack Name	Date	Attack Start Time	Attack Finish Time
172.31.70.4 (Valid IP:18.221.219.4)	172.31.69.25 (Valid IP:18.217.21.148)	FTP-BruteForce	Wed-14-02-2018	10:32	12:09
172.31.70.6 (Valid IP:13.58.98.64)	18.217.21.148-172.31.69.25	SSH-Bruteforce	Wed-14-02-2018	14:01	15:31
172.31.70.46 (Valid IP:18.219.211.138)	18.217.21.148-172.31.69.25	DoS-GoldenEye	Thurs-15-02-2018	9:26	10:09
172.31.70.8 (Vazlid IP:18.217.165.70)	18.217.21.148-172.31.69.25	DoS-Slowloris	Thurs-15-02-2018	10:59	11:40

	How to incorporate new attack cases
XGB	Retrain by integrating them into the existing dataset
TabPFN	**Reconstruct the context** within the pre-trained weights

v3 architecture

① Encode inputs · Cell embedding

Converts values such as packet length into vectors
Also indicates which entries are missing

② Summarize columns · Feature embedding

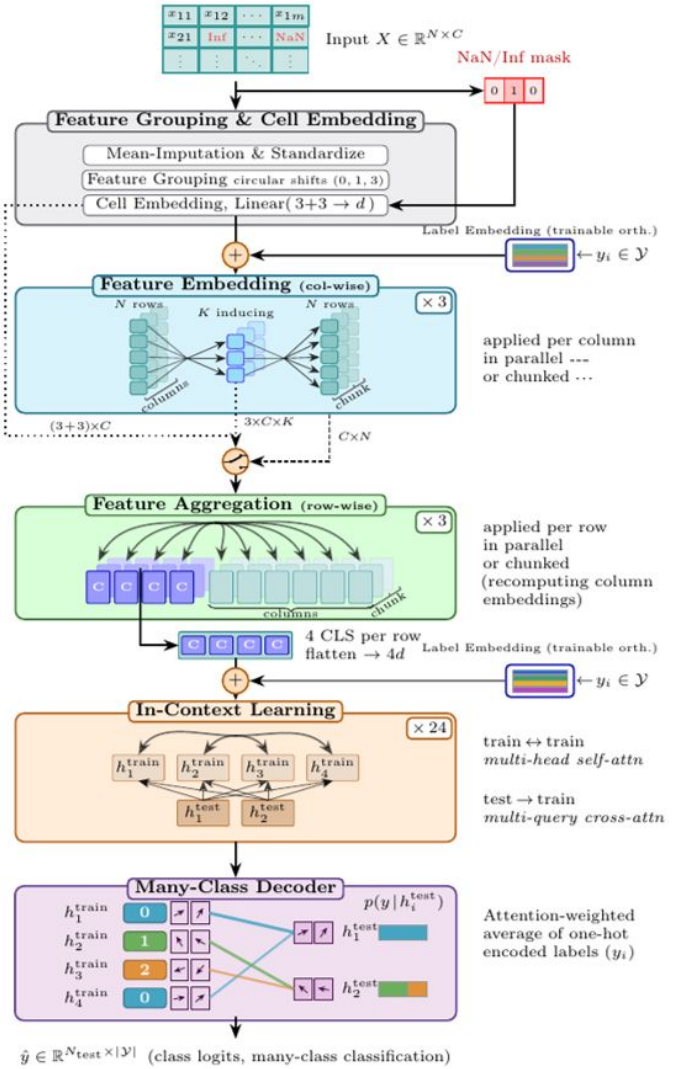
Summarizes each feature across multiple flows

Uses the benign and attack labels in the context

③ Combine features · Feature aggregation

Combines each flow's features into one representation

Uses it to compare relationships between flows



v3 architecture

④ Consult examples · In-context learning

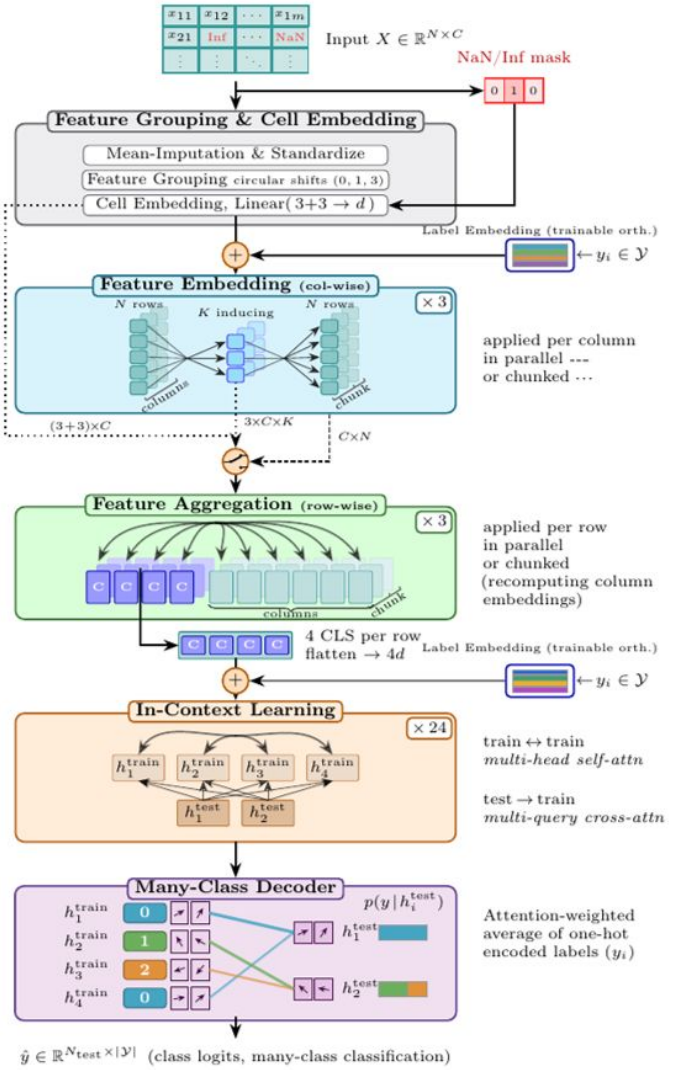
A new flow queries the labeled context examples

Attention identifies relevant examples for prediction

⑤ Predict probabilities · Many-class decoder

Assigns greater weight to labels of relevant examples
Combines them into benign and attack probabilities

Keep context examples of existing classes to continue predicting those classes



Evaluation scenario

By providing a series of attack cases in stages, comparing the update costs and detection quality of the two models.

CIC2018

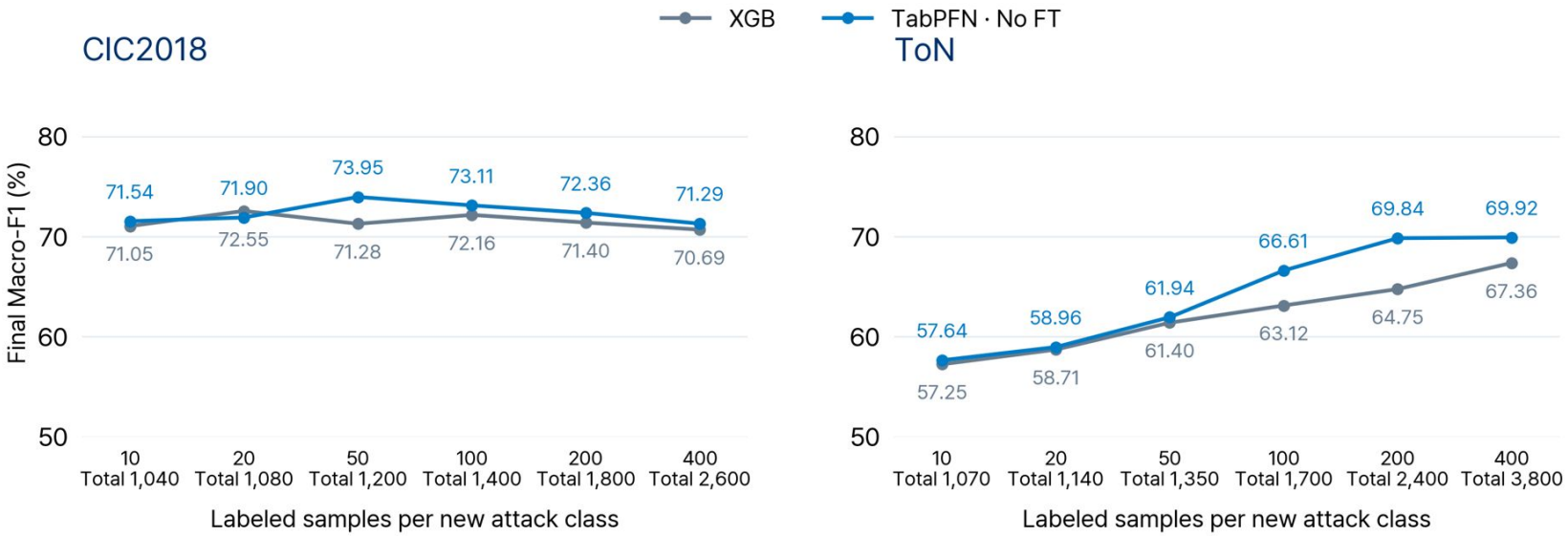
Stage	New classes	Per class	Cumulative
1	Benign Brute force DoS	750 125 125	1,000
2	DDoS	100	1,100
3	Web attacks	100	1,200
4	Infiltration	100	1,300
5	Bot	100	1,400

ToN

Stage	New classes	Per class	Cumulative
1	Benign DoS Scanning	750 125 125	1,000
2	DDoS Injection	100 100	1,200
3	Password	100	1,300
4	XSS	100	1,400
5	Backdoor Ransomware	100 100	1,600
6	MitM	100	1,700

Performance by new attack sample budget

Initial 1,000 fixed · 10/20/50/100/200/400 per new attack class · Final model, full test



400 per new attack · TabPFN minus XGB: CIC2018 +0.60 pp · ToN +2.56 pp

Results

Final model, full test · XGB: cumulative retraining · TabPFN: cumulative context, no FT

